



From visuals to value: leveraging generative AI to explore the economic implications of movie poster

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ABSTRACT

This study introduces a novel exploration into the impact of visual elements on consumer behavior in the film industry, utilizing generative AI. By employing expectancy violations theory, this study examines how the mood and tone suggested by movie posters—a key visual advertising tool—contrast with the actual content of the films, revealing a significant influence on consumer decisions. Through feature extraction from movie posters using generative AI and regression model analysis, this study demonstrates that deviations from audience expectations, as suggested by movie posters, positively affect box office performance. However, this impact varies depending on whether the movie is produced by major or non-major studios, with major studio productions benefiting more from mood and tone congruence. This study extends existing literature by highlighting the role of visual cues in movie posters and offers practical insights for movie marketers using expectancy violations to enhance audience engagement and box office performance.

1. Introduction

Visual advertising utilizes the power of imagery, color, and design to capture attention and convey messages swiftly and effectively. In the competitive landscape of consumer marketing, visual elements play a pivotal role, guiding potential customers' perceptions and influencing their decision-making processes (Burns et al., 1993; Heitmann et al., 2020; Jia et al., 2014; Wang et al., 2020). These elements are not just aesthetic enhancements but strategic tools that convey products' intrinsic values and emotional appeals, making them particularly crucial in markets where consumer choices are highly influenced by visual impressions.

The significance of visual elements in advertising extends across a spectrum of industries, but it is particularly impactful in the marketing of experiential goods (Karray & Debernitz, 2017; Septianto et al., 2021; Wei & Lu, 2013). Experiential goods such as films, live concerts, and travel experiences offer intangible benefits that are often difficult to articulate through words alone. Unlike tangible products, experiential goods cannot be evaluated prior to purchase, so visuals act as a bridge, providing potential consumers with a preview of the experience, setting expectations, and enticing emotional engagement before the actual purchase is made. For instance, the visual design of a movie poster,

which is one of the essential promotional media, can hint at the film's narrative complexity or emotional depth, potentially swaying a consumer's decision to watch it.

Especially in the context of film marketing, the strategic importance of movie posters cannot be overstated. These posters are often the first point of visual contact between the movie and its potential audience, and they carry the substantial task of encapsulating the essence of the film in a single frame. Traditionally, movie posters were prominently displayed as visual aids at physical locations such as theaters and billboards. However, in today's digital environment they have evolved to become crucial components of online marketing strategies, often appearing as thumbnails on streaming platforms and social media. These digital thumbnails are designed to spark curiosity and capture the interest of potential viewers.

Despite their importance, studies on the impact of visual elements in movie posters remain under-explored in the marketing literature. Previous research in this field has only examined the role of movie posters in predicting consumer perceptions (Chen et al., 2022), interests (Matsuzaki et al., 2017), and genres (Ivasic-Kos et al., 2015). In this study, we aim to examine the visual element of movie posters to uncover how they can influence movie viewer perceptions, drive interest, and ultimately impact a film's financial success, providing new insights into

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the role of visual marketing in the film industry.

Specifically, we aim to understand how audiences' perceptions of inconsistencies between movie posters and textual information influence their response. Expectancy violations theory (Burgoon, 1993) suggests that when there is a mismatch (incongruencies) between what viewers expect from movie posters and the actual movie content, audience interest increases. Such expectancy violations have been investigated in computer-mediated communication contexts, such as Facebook unfriending (Bevan et al., 2014; McLaughlin & Vitak, 2012). However, there is no study in the marketing literature that examines such incongruencies, particularly in the context of movies. Therefore, this study extends the line of research to understand the importance of incongruencies between movie posters and release information, as a contextual factor, in box office performance.

As artificial intelligence has developed to the human level, text or image studies related to machine learning increased (Azadi et al., 2023; Hermann & Puntoni, 2024; Huh, Nelson, & Russell, 2023; Wang et al., 2024a), presenting an opportune area to explore. To extract features from movie posters, we employed a generative AI technique, a powerful model driven by large language models that is capable of interpreting images (Team, 2023). This model helped us to extract descriptive adjectives depicting the moods and tones conveyed by movie posters. Then, we compared these extracted tones to the ones in published movie information such as movie summaries, using text-mining techniques and cosine similarity measures (Zhelezniak et al., 2019). Finally, we applied regression analysis to assess how such contrasts between movie posters and textual released information affect box office performance.

Our findings suggest that expectancy violation in visual elements can indeed intensify audience interest, which in turn affect consumer decision making. These findings provide novel perspectives to the understanding of consumer purchasing behavior for experiential goods and extend the expectancy violations theory to a new application in the literature. From a methodological perspective, this study introduces a state-of-the-art technique for examining emotional representations in images using generative AI. From a practical perspective, our findings offer strategic implications for movie marketers, emphasizing the importance of visual elements in promotion and the utilization of advanced techniques for better prediction.

The study is organized as follows. We first review the relevant literature on movie posters and expectancy violations theory. Next, we detail our research methodology. The subsequent section presents our findings. We conclude by discussing the theoretical and practical contributions of this study and its limitations.

2. Literature review

2.1. The role of movie posters

Movie posters serve as a critical marketing tool in the film industry, summarizing the fundamental story of a film and conveying it to potential audiences (Sun et al., 2021). Visual elements are recognized as crucial components in conveying messages and evoking emotional responses (Bashirzadeh et al., 2022; Feld et al., 2013; Rumpf et al., 2020). In this regard, movie posters stand out as dynamic visual representations that leverage images, colors, typography, and design elements to capture attention and convey key themes and emotions associated with the film (Wi et al., 2020). Moreover, by harnessing the principles of visual communication, movie posters effectively engage viewers and stimulate interest in the movie, ultimately influencing the decision-making process of moviegoers (Gazley et al., 2011).

In the context of marketing, the role of movie posters extends beyond mere promotional material (Stokmans, 2015). They are considered strategic assets in the film marketing mix. Scholars argue that movie posters function as persuasive communication tools, employing visual and textual elements to evoke emotions, convey themes, and create memorable impressions (Chen et al., 2022; Chu & Guo, 2017; Sun et al.,

2021). The visual components of movie posters, including color schemes, imagery, mood, and typography, are meticulously crafted to resonate with target audiences and stimulate interest in the film (Mehrabian, 2011). Thus, movie posters serve as tangible representations of a film's brand identity, contributing to its overall market positioning and differentiation in a crowded entertainment landscape.

Furthermore, the advent of digital marketing platforms and social media has amplified the importance of advertisements, such as movie posters and trailers, in contemporary film promotion strategies (Hu et al., 2023). The visual contents on online platforms influence the cognitive, affective, and behavioral dimensions of engagement (Bazi et al., 2023), highlighting how movie posters serve as versatile assets that can be adapted for various digital formats, including social media posts, website banners, and email campaigns. This dynamic digital area presents new opportunities for filmmakers and marketers to experiment with innovative poster designs and engage with audiences on multiple platforms, thereby amplifying the reach and impact of their promotional efforts.

Despite their significance, empirical research examining the impact of movie posters on box office revenue remains limited within marketing literature. However, studies in related fields offer insights into the potential influence of movie posters on consumer perceptions and preferences (Chen et al., 2022). For example, scholars have examined the role of movie posters in predicting genres and moods (Ivasic-Kos et al., 2015; Sung & Chokshi, 2017), furthermore predicting consumer interests (Matsuzaki et al., 2017). Applying these principles to movie marketing, it can be inferred that compelling and visually appealing movie posters have the potential to attract larger audiences and drive box office success.

Therefore, understanding the impact of movie posters on consumer behavior and box office performance becomes increasingly crucial from both academic and practical perspectives, aiming to maximize the success of cinematic endeavors. However, to the best of our knowledge, there is no study addressing this issue. To fill this gap, we endeavor to address the role of movie posters in shaping consumer perceptions and driving box office revenue, thereby expanding the scope of marketing research.

2.2. Expectancy violations theory

Expectancy violations theory offers valuable insights into how individuals interpret and respond to unexpected or unconventional stimuli (Burgoon, 1978; Guerrero et al., 2017). According to this theory, individuals develop expectations based on social norms, cultural cues, and past experiences, then deviations from these expectations can lead to heightened attention, arousal, and cognitive processing. When these expectations are violated, either positively or negatively, individuals experience a state of arousal that prompts them to reassess their perceptions and adjust their behavior accordingly (Burgoon, 2015).

The heart of the theory lies in the concept of expectancy, which refers to the prediction that individuals make about how others will behave in a given situation (Burgoon, 1993). These expectations serve as mental frameworks that guide social interactions and help individuals navigate interpersonal relationships (Afifi & Metts, 1998). For instance, when an individual's behavior deviates from such expectations, it creates a violation, triggering a cognitive response as individuals attempt to make sense of the unexpected behavior and reconcile it with their preconceived notions.

Expectancy violations theory proposes that the evaluation of expectancy violations involves two primary factors: violation valence and violation magnitude (Afifi & Metts, 1998). Violation valence refers to the perceived positive or negative nature of the deviation from expectations, while violation magnitude relates to the extent to which the violation exceeds or falls short of expectations (Bevan et al., 2014). For instance, positive violations occur when the unexpected behavior is perceived as favorable or rewarding, whereas negative violations

involve behaviors that are perceived as undesirable or off-putting.

Furthermore, the theory distinguishes between personal and contextual factors that influence individuals' reactions to expectancy violations (Burgoon, 1993). Personal factors, such as personality traits, cultural background, and relational history, shape individuals' baseline expectations and their tolerance for deviation from these expectations (Bevan, 2003; Cohen, 2010). Contextual factors, including situational dynamics, social norms, and relational context, also play a role in determining how expectancy violations are perceived and interpreted (Burgoon & Jones, 1976). In this research, we focus on the impact of contextual factors that violate individuals' expectations.

Expectancy violations theory has been applied to various contexts beyond interpersonal communication, including marketing, and media studies (Azer et al., 2023; Chen et al., 2022; Darke et al., 2010; Sun et al., 2024). In these contexts, expectancy violations can manifest through unexpected or unconventional messaging strategies, visual elements, or narrative techniques (Bevan et al., 2014). By strategically leveraging expectancy violations, marketers can capture audience attention, stimulate curiosity, and create memorable brand experiences that stand out in a crowded marketplace (Sun et al., 2024). Overall, expectancy violations theory offers valuable insights into how individuals interpret and respond to deviations from their expectations, shedding light on the complexities of human communication and behavior in diverse social and cultural contexts. In this study, we apply this theory to the film industry to examine the impact of movie posters on box office revenue when those posters violate potential customers' expectations.

2.3. Hypothesis development

2.3.1. Impact of mood and tone on box office revenue

Expectancy violations theory posits that individuals develop expectations based on past experiences and cultural norms, and deviations from these expectations can lead to high attention (Burgoon, 1993). In the context of the film industry, when the elements depicted in a movie poster closely align with the release information of the movie, it may not capture audience interest as effectively as when there is an unexpected deviation from the information. This congruence can create negative expectancy violations, wherein audiences perceive the film as predictable or lacking novelty. In this research, we empirically compare two elements between movie posters and the released film information: the mood and tone of the posters against the genre and synopsis of the films.

Theoretically, this research proposes that the congruence between the elements depicted in the poster and the movie's descriptive information creates negative expectancy violations, as they fail to challenge or intrigue potential moviegoers. For instance, a romantic comedy film's poster that showcases dark, ominous imagery instead of the typical light-hearted and romantic theme, results in a negative expectancy violation due to the disparity with the genre's anticipated imagery.

Furthermore, we measure the impact of the congruence between the elements of movie posters and movie information on box office revenue. The deviations from movie information norms can lead to positive expectancy violations, enhancing audience engagement and ultimately driving box office success. Thus, by considering the impact of expectancy violations on audience behavior, box office forecasting models can better anticipate box office performance and assist filmmakers and marketers in optimizing their promotional strategies. In other words, we propose a novel variable that can be significant in predicting box office revenues. Thus, we hypothesize the following:

H1. A high degree of similarity between the poster mood and movie genre negatively affects the film's opening week box office revenue.

H2. A high degree of similarity between the tones of the poster and the movie synopsis negatively affects the film's opening week box office revenue.

2.3.2. Moderating effect of major studio

Major studios, with their established reputations and substantial marketing resources, shape audiences' expectations and reactions in

ways that differ significantly from those influenced by independent studios (Elberse & Eliashberg, 2003; Elberse, 2007; Joshi & Hanssens, 2009; Liu et al., 2015). For instance, Elberse and Eliashberg (2003) confirmed that major studios significantly impact the initial screen allocation for movies, leading to higher revenues in the opening week. Liu et al. (2015) also found that major studios can significantly enhance the release scale and opening week revenues. These studies emphasize the crucial role of major studio's marketing distribution capabilities in the success of movies.

Therefore, this study also considers film studio as a moderating factor in understanding the dynamics influencing box office revenue, particularly in instances where consumers possess limited prior information about the movie. Films produced by major studios, backed by powerful marketing efforts, are more likely to be exposed to consumers before release. This increased exposure leads to heightened consumer expectations. Thus, meeting these expectations becomes more crucial for major studios compared to non-major studios. So, we propose following hypotheses:

H3. For major studios, a high degree of similarity between the poster mood and movie genre positively affects the film's opening week box office revenue.

H4. For major studios, a high degree of similarity between the tones of poster and movie synopsis positively affects the film's opening week box office revenue.

3. Method

3.1. Data sources

To examine our hypotheses, we collected data from various movie review platforms. We employed Python packages, such as Selenium and BeautifulSoup, for data scraping purposes. Table 1 provides an overview of the variables and their respective data sources. Our dataset consists of 2,918 movies selected based on three criteria: (1) movies released in the United States between 2012 and 2016, (2) availability of domestic revenue data on the first week of release, and (3) accessibility of a movie poster.

We mainly obtained the movie information from three primary sources. From The-numbers.com, we collected box office revenues, the number of screens, production countries, movie genres, release date, and the Motion Picture Association of America (MPAA). In order to select a representative movie genre among multiple genres of a movie, we collected genre data from the-numbers.com, following Liu et al. (2015). This study employs a single major genre for each movie, consistent with approaches used in prior research (Dellarocas et al., 2007; Liu et al., 2015; Moon et al., 2010). Then, we defined the main movie poster based on two criteria: (1) whether the thumbnails from two popular websites (Rottentomatoes.com and IMDB.com) match, and (2) if they do not match, the majority of posters sharing the same image. There are multiple movie posters in IMDB.com that share the same images but are presented in different languages. We excluded the movie posters that contained provocative or sexual content due to model input restrictions.¹ Lastly, we collected the numbers of critic reviews and ratings, and user review ratings and numbers from IMDB.com. Fig. 1 outlines the steps of our measurement process for mood and tone.

3.2. Similarity of movie poster and genre detecting mood from the image

Historically, detecting moods from images has posed a challenge due to their inherent embodiment of sentiments. However, leveraging the advancements in large language models and generative AI, this study

¹ Gemini API blocks harmful content, such as content that violates the prohibited use policy, including sexually explicit. Therefore, those images affected by the BlockedPromptException error could not be utilized.

Table 1
Variables and descriptions.

Variable	Description	Data source
<i>DVs</i>		
lnFirstRevenue	The log-transformed box office revenue for opening week.	the-numbers.com
lnRevenue	The log-transformed total box office revenue.	the-numbers.com
<i>Movie-related IVs</i>		
Mood&Genre	How similar the mood is depicted in the movie poster to the genre of the movie.	Calculated based on data from rottentomatoes.com and the-numbers.com
Tone	Whether the tone of the movie poster is matched with the tone of synopsis.	
lnScreen	The log transformed total number of theaters in the opening week.	the-numbers.com
lnMaxScreen	The log transformed max number of theaters during the period.	the-numbers.com
lnActorPower	The log-transformed sum of the aggregate box office revenues of the movies in which the leading actors were involved.	the-numbers.com
CriticRating	The percentage of approved tomatometer critics who have given this movie a positive review.	rottentomatoes.com
lnCriticCount	The log-transformed number of approved tomatometer critic reviews.	rottentomatoes.com
ReviewRating	The audience review rates.	imdb.com
lnReviewCount	The log-transformed number of viewer reviews for movies.	imdb.com
Country	The country the movie is produced: United States, Foreign, Combined	the-numbers.com
Genre	The genre of the movie: Drama, Documentary, Comedy, Thriller/Suspense, Action, Adventure, Romantic/Comedy, Horror, ETC	the-numbers.com
Year	The year the movie is released.	the-numbers.com
Month	The month the movie is released.	the-numbers.com
MajorStudio	Whether the movie is released by a major studio	https://en.wikipedia.org/wiki/Major_film_studios
ViewingRating	The MPAA rating system categorizes movies by age-appropriateness: G, NC-17, PG, PG-13, R, Not Rated	the-numbers.com

aims to introduce a pioneering application of such models. Specifically, we aim to extract moods, adjectives, or emotional tones conveyed by images (movie posters) through innovative computational techniques. This approach promises to enhance our understanding of the emotional content embedded within visual media, thereby opening avenues for more sophisticated analysis and interpretation in various domains such as marketing.

3.2.1. Image preprocessing

This study aims to extract moods from images, specifically excluding any text present within these images. To achieve this, we first removed all text from the poster images. This was accomplished through a two-step process: initially identifying the text using optical character recognition (OCR), and then erasing it with an inpainting algorithm. For text recognition in images, we utilized Keras-ocr, a comprehensive package that integrates the capabilities of the Character Region Awareness for Text Detection (CRAFT) model (Baek et al., 2019) and the Convolutional Recurrent Neural Network (CRNN) model (Shi et al., 2016). The CRAFT model detects text areas within images, while the CRNN model is responsible for recognizing the detected text via feature extraction. Keras-ocr outputs the coordinates (x, y) of the bounding box

corners for each detected word. With the coordinates, the OpenCV inpainting algorithm (Bertalmio et al., 2001) removed the texts smoothly. The algorithm is based on fluid dynamics principles and uses partial differential equations. It starts by mapping out from known to unknown areas. It follows isophotes, lines linking points of equal brightness, akin to elevation contour lines. The algorithm ensures the alignment of gradient vectors around the area to be filled in. Once aligned, it applies color to achieve a consistent look.

3.2.2. Mood extraction

Previous literature has utilized advanced AI models to interpret the mood conveyed by images for accurate sentiment analysis in visual media (Chao et al., 2023; Zhou et al., 2019). As a result, these models can produce more detailed and scalable evaluations of emotional content from images. However, a significant challenge arises because emotions in images are multi-dimensional. This means that a single image can be described using multiple adjectives. To address this issue, studies in computer science and multimedia have employed adjectives to characterize the emotional content from images, and then combine these adjectives to extract the representative emotions (Borth et al., 2013; Das & Singh, 2023; Jou et al., 2015; Wang et al., 2016). Following this approach, this study also employed the same method to extract mood descriptors.

In order to extract mood from the movie poster image, we utilized Gemini-pro (Team, 2023), which is a state-of-the-art technique developed by Google. Gemini is a family of powerful and versatile multimodal models driven by large language models that can understand and operate across various types of information, including images and texts. Its capabilities extend beyond traditional language models, making it a significant advancement in the field of artificial intelligence. The model is built upon Transformer decoders (Vaswani et al., 2017), which has been optimized for greater efficiency and scalability. These modifications ensure stable training scalability and more efficient inference. The model not only creates image captions but also answers questions considering the image. In this study, we used prompts to obtain the mood of the poster image. Fig. 2 shows the structure of Gemini for mood and tone extraction. To enhance the precision of the model outputs, we requested the creation of a list comprising five adjectives to describe the mood conveyed in the prompts. Jou et al. (2015) demonstrated that using top five words accuracies significantly improved performance compared to top one-word accuracies in capturing the range of emotions associated with images. Thus, to ensure a comprehensive description of the mood of a poster, we selected five words to represent its emotional content. The prompt given was, "What's the mood to describe the image, in a list of five adjective words," with the preprocessed poster images. The examples of the posters and mood can be found in Appendix A.

3.3. Mood and genre similarity measure

The similarity between the adjectives of moods and a genre is measured by using cosine similarity (Zhelezniak et al., 2019). We utilized pre-trained Word2Vec² vectors (Church, 2017) to convert the extracted moods and genres into numerical representations. Subsequently, cosine similarity was employed to quantify the similarity between the mood adjectives and the genre for each movie. The cosine similarity is calculated as:

$$\text{Cosine Similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

² We have utilized Word2Vec Google News 300, which provides 300-dimensional vectors for approximately 3 million words and phrases. These pre-trained embeddings allow for the quantification of semantic similarity by measuring cosine similarity between word vectors in the high-dimensional space.

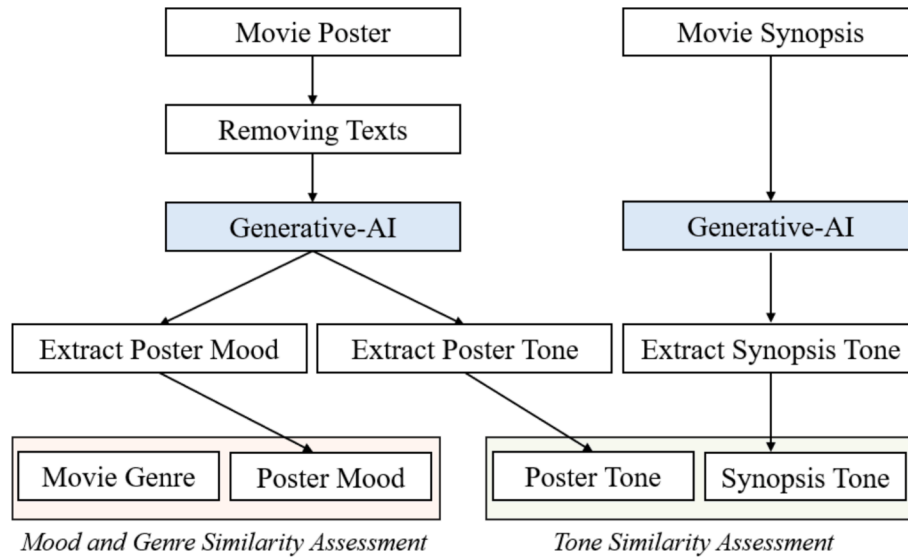


Fig. 1. Process of measurements.

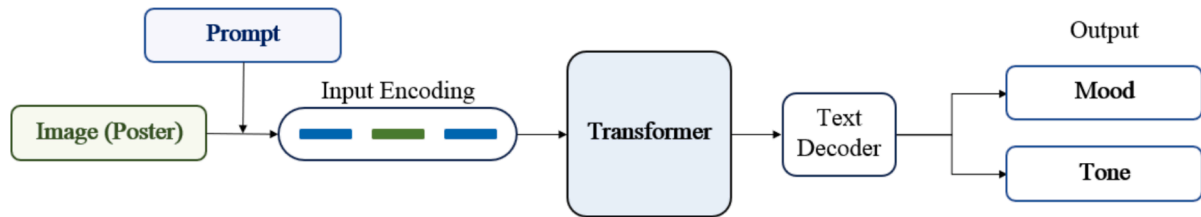


Fig. 2. Structure of Gemini for mood and tone extraction.

where $A \cdot B$ represents the dot production of vectors A and B , and $\|A\| \|B\|$ represent the magnitudes of vectors A and B . In this context, vector A is a numerical representation of a mood adjective. This vector is obtained through a word embedding model, Word2Vec, which translates the adjective into a high-dimensional space where each dimension captures a different semantic or syntactic property of the word. Vector B corresponds to the numerical representation of a movie genre, derived in a similar manner to encapsulate the defining features of the genre within the same vector space. The cosine similarity, therefore, provides a measure of the orientation, or the directional agreement, between the mood descriptor and the genre representation, indicating the degree of their semantic relatedness. Then, the mean value of the cosine similarities was calculated for each movie:

$$\text{Mood\&Genre} = \frac{1}{5} \sum_{i=1}^5 \text{Cosine Similarity}(A_i, G)$$

where $\text{Cosine Similarity}(A_i, G)$ represents the cosine similarity between the i^{th} adjective vector A_i and the genre vector G and $\frac{1}{5}$ averages these values. Given that a single image may evoke multiple moods, and determining a dominant one is often ambiguous, an average was computed to provide a more balanced representation. This average offers a view of the mood-genre relationship, providing a numerical representation of the relationship.

3.3.1. Importance of mood and genre similarity

To evaluate the feature importance, we employed two machine learning techniques. First, we investigated the feature importance by the random forest model (Wang et al., 2024b). Second, we employed the

explainable AI, SHapley Additive exPlanations (SHAP), to interpret the result of XGBoost model. (Meng et al., 2020; Wang et al., 2024b). The feature importance of Mood&Genre ranked in the top five, outperforming genre and major studio, highlighting the significance of visual elements. In SHAP, the mean absolute value of SHAP values represents a feature's average impact on model predictions across all samples. It considers both positive and negative contributions equally, making it a clear measure of feature importance. This value is often used to rank features in global importance plots. The results show that Mood&Genre, representing mood and genre similarity, is the top 5 important features in both models (please refer to Fig. 3 and Fig. 4). This means that Mood&Genre has one of the highest average impacts on model predictions across all samples, compared to other variables.

3.3.2. Alternative variable – Tone similarity

We also consider another factor that contributes to the overall aesthetic and emotional impact of an image, tone. While mood primarily focuses on the emotional content of the image, tone is more concerned with the distribution of light and shadow. We categorized the tone of movie posters into three: bright, dark, and neutral. We also applied Gemini to extract the tones of the movie poster. For instance, we used a prompt such as: "What's the tone to describe the image, in one word: 'bright,' 'dark,' or 'neutral'." To compare the tones conveyed from movie posters from the released information, we also extracted the tone from the movie summary (synopses) by Gemini with the following prompt: "What's the tone to describe the following story, in one word: 'bright,' 'dark,' or 'neutral'." We examined whether these two tones from movie poster and synopsis matched or not and defined this as tone

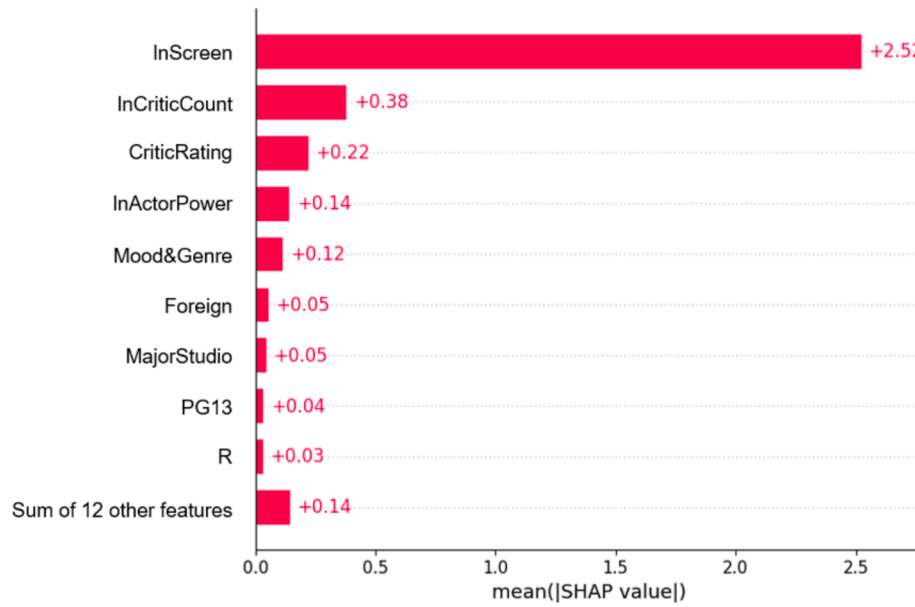


Fig. 3. Results of SHAP with XGBoost.

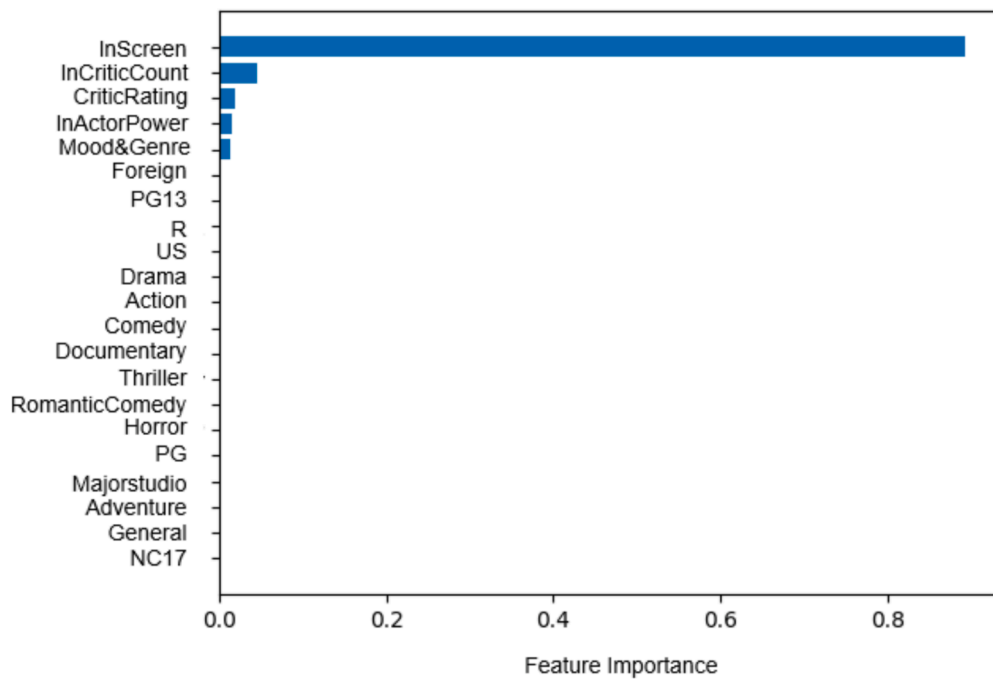


Fig. 4. Results of feature importance with random forest.

similarity. In Table 2, we present the top 10 words from mood associated with two contrasting tones. These consistent outputs of the model show that the tones accurately represent the mood described by the words. In addition, three coders, who have professional knowledge in this research field, validated the outputs of the model using a sample of 500 movies, achieving an agreement rate of 89.06 %.

3.4. Variable description

The dependent variables in our study are represented by domestic box office revenues in the US market. We focus on opening week revenue, as our primary focus is to analyze the impact of the movie poster before any public feedback or wider audience reactions become

available (Akdeniz & Talay, 2013; De Vany and Walls, 1999; Elberse & Eliashberg, 2003; Joshi & Mao, 2012; Liu et al., 2015; Liu, 2006). For the independent variables, we first assess the similarity between the mood of the image and the movie genre through similarity measurement (called “Mood&Genre” in this study). Then, we use a binary variable indicating alignment between the image’s tone and the movie synopsis’s tone (called “Tone” in this study). The model incorporated control variables that have been examined in previous studies (Akdeniz & Talay, 2013; Bae & Kim, 2019; Liu et al., 2015; Liu, 2006), including screen count, star power, critic reviews, and their frequencies, audience reviews and their volumes, the country of production, genre, the movie’s release timing (year and month), major studio, and viewing age. The summary statistics of the variables in this study are presented in Tables 3

Table 2

Frequency of the words of the mood for each tone.

Bright		Dark	
Words	Frequency	Words	Frequency
happy	123	mysterious	260
romantic	122	intense	240
playful	114	suspenseful	198
long	109	eerie	197
nostalgic	92	dramatic	122
energetic	69	dark	117
funny	60	moody	95
carefree	58	nostalgic	78
peaceful	53	determined	77
heartwarming	51	desolate	76

Table 3

Summary statistics of variables.

Variable	Observations	Mean	Std. Dev.	Min	Max
<i>DVs</i>					
lnFirstRevenue	2,918	11.28	3.30	3.22	19.33
lnRevenue	2,918	12.85	3.32	4.29	20.66
<i>Movie-related IVs</i>					
Mood&Genre	2,903	0.17	0.10	-0.05	0.56
Tone	2,918	0.48	0.50	0	1
lnScreen	2,918	3.31	2.76	0.69	8.39
lnMaxScreen	2,918	4.17	2.52	0.69	8.39
lnActorPower	2,918	17.59	6.48	0	24.81
CriticRating	2,866	62.74	26.92	0	100
lnCriticCount	2,918	102.56	152.85	0	3,681
ReviewRating	2,918	6.45	1.03	1.30	9.30
lnReviewCount	2,918	9.84	2.34	2.30	17.18

Table 4

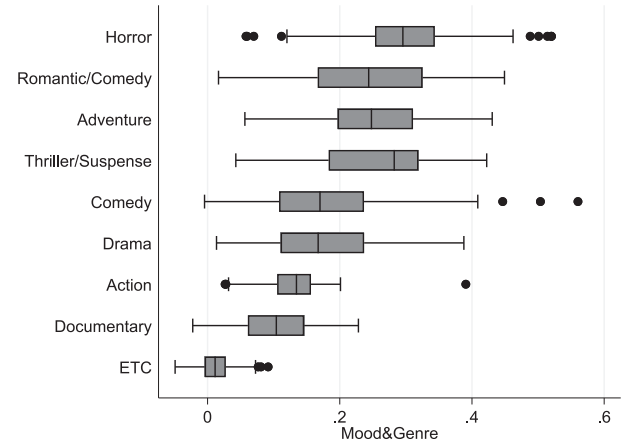
Summary statistics of categorical variables.

Type	Proportion
Tone	Matched (48.39 %), Not-Matched (51.61 %)
Country	United States (57.78 %), Foreign (34.34 %), Combined (7.88 %)
Genre	Drama (33.10 %), Documentary (17.79 %), Comedy (13.57 %), Thriller/Suspense (9.42 %), Action (7.75 %), Adventure (5.83 %), Romantic/Comedy (4.42 %), Horror (4.39 %), ETC (3.74 %)
Year	2012 (19.26 %), 2013 (19.50 %), 2014 (20.49 %), 2015 (19.53 %), 2016 (21.21 %)
Month	Jan (6.07 %), Feb (6.61 %), Mar (9.25 %), Apr (9.36 %), May (8.29 %), Jun (8.36 %), Jul (7.54 %), Aug (9.18 %), Sep (10.83 %), Oct (10.14 %), Nov (8.09 %), Dec (6.27 %)
MajorStudio	MajorStudio (12.61 %), Others (87.39 %)
Viewing Rating	R (32.69 %), PG-13 (20.56 %), PG (7.40 %), G (0.55 %), NC-17 (0.10 %), Not Rated (38.69 %)

and 4.

3.5. Descriptive study

Fig. 5 provides a comparative view of movie genres against the ‘Mood&Genre’ index, a measure of the similarity between the mood of movie posters and the movie genre. ‘Horror’ exhibits a high level of consistency, suggesting a close alignment between its poster mood and genre expectations. In contrast, ‘Romantic/Comedy’ spans a wide range, indicating varied levels of harmony or disparity in mood and genre. ‘Adventure’ and ‘Thriller/Suspense’ genres have similar median ‘Mood&Genre’ scores, with ‘Thriller/Suspense’ having outliers that potentially represent movies with a particularly strong or weak alignment. ‘Comedy’ and ‘Drama’ show lower median values, possibly reflecting a less consistent genre-to-mood match. A notable outlier in the ‘Action’ genre suggests an exceptional case where the movie poster mood significantly differs from the genre. ‘Documentary’ and ‘ETC’ hold

**Fig. 5.** The average of Mood&Genre for each genre.

the lowest median scores, with ‘ETC’ exhibiting minimal variation, which could point to a more generalized or ambiguous genre representation. The plot’s analysis underscores the varying degrees of alignment between movie poster moods and their respective genres, which can profoundly affect audience expectations and engagement. These insights are crucial for movie producers and marketers in creating and promoting content that resonates accurately with its intended audience.

3.6. Model

The following regression model examines the main hypotheses outlined in this study:

$$\ln \text{FirstRevenue}_i = \beta_0 + \beta_1 \text{Mood\&Genre}_i + \beta_2 \ln \text{Screen}_i + \beta_3 \ln \text{ActorPower}_i + \beta_4 \text{CriticRating}_i + \beta_5 \ln \text{CriticCount}_i + \beta_5 \text{Country}_i + \beta_6 \text{Genre}_i + \beta_7 \text{Year}_i + \beta_8 \text{Month}_i + \beta_8 \text{MajorStudio}_i + \beta_9 \text{ViewingRating}_i + \epsilon_i \quad (1)$$

$$\ln \text{FirstRevenue}_i = \beta_0 + \beta_1 \text{Tone}_i + \beta_2 \ln \text{Screen}_i + \beta_3 \ln \text{ActorPower}_i + \beta_4 \text{CriticRating}_i + \beta_5 \ln \text{CriticCount}_i + \beta_5 \text{Country}_i + \beta_6 \text{Genre}_i + \beta_7 \text{Year}_i + \beta_8 \text{Month}_i + \beta_8 \text{MajorStudio}_i + \beta_9 \text{ViewingRating}_i + \epsilon_i \quad (2)$$

$$\ln \text{FirstRevenue}_i = \beta_0 + \beta_1 \text{Mood\&Genre}_i + \beta_2 \text{Tone}_i + \beta_3 \text{Mood\&Genre}_i \times \text{MajorStudio}_i + \beta_4 \text{Tone}_i \times \text{MajorStudio}_i + \beta_5 \text{MajorStudio}_i + \beta_6 \ln \text{Screen}_i + \beta_7 \ln \text{ActorPower}_i + \beta_8 \text{CriticRating}_i + \beta_9 \ln \text{CriticCount}_i + \beta_{10} \text{Country}_i + \beta_{11} \text{Genre}_i + \beta_{12} \text{Year}_i + \beta_{13} \text{Month}_i + \beta_{14} \text{MajorStudio}_i + \beta_{15} \text{ViewingRating}_i + \epsilon_i \quad (3)$$

where i refers to a movie. The model was estimated using ordinary least squares. To examine the impact of mood and genre similarity on box office revenue of the opening week, we have considered the variables that would influence consumers before the movie’s release. So, we have not included the max screen numbers and audience reviews. However, these factors will be included in later phases of our research to offer a comprehensive understanding of their impact on movie revenue.

4. Results

4.1. Main results

Table 5 provides our main results. The findings support Hypothesis 1, as evidenced by the negative and statistically significant coefficient for

Table 5

Linear regression on the opening week box office revenue.

VARIABLES	(1) lnFirstRevenue	(2) lnFirstRevenue
Mood&Genre	−0.475* (0.262)	
Tone		−0.113*** (0.041)
lnScreen	1.040*** (0.012)	1.041*** (0.012)
lnActorPower	0.024*** (0.004)	0.023*** (0.004)
CriticRating	0.014*** (0.001)	0.014*** (0.001)
lnCriticCount	0.374*** (0.028)	0.381*** (0.028)
Genre		
Action	0.195 (0.128)	0.146 (0.125)
Adventure	0.240 (0.147)	0.117 (0.131)
Comedy	0.120 (0.122)	0.009 (0.115)
Documentary	0.158 (0.117)	0.090 (0.113)
Drama	0.189* (0.113)	0.090 (0.105)
Horror	−0.244 (0.156)	−0.341** (0.139)
RomanticComedy	0.222 (0.155)	0.077 (0.142)
Thriller	−0.010 (0.133)	−0.094 (0.118)
MajorStudio	0.341*** (0.063)	0.339*** (0.063)
Constant	5.140*** (0.178)	5.166*** (0.178)
ViewingRating	YES	YES
Country	YES	YES
Year	YES	YES
Month	YES	YES
Observations	2,851	2,866
R-squared	0.907	0.907

Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.10.

Mood&Genre ($\beta_1 = -0.475$, $p < 0.10$). This finding indicates that the similarity in the mood of the poster and the movie genre has a negative impact on the box office revenues in its opening week. In other words, the movies with opposite poster mood and movie genre are likely to have higher revenues for opening week. In Column (2) of Table 5, the coefficient of Tone ($\beta_1 = -0.113$, $p < 0.01$) also had significant negative effects on the opening week box office revenues, supporting Hypothesis 2. This finding indicates that the dissonance between the tones of movie posters and synopses increases the opening week box office revenues. The two variables showed the same significant effects on the opening week revenues.

4.2. Interaction effect

We examined whether the effects of mood and tone interact with major studios. To account for ‘MajorStudio,’ which refers to the wider distribution of movies produced by major studios, we collected the names of the distributors. We used a dummy variable to indicate whether a movie was distributed by a major studio, following the approach of Elberse and Eliashberg (2003). According to the results in Table 6, the movies produced by major studios with a high similarity between poster mood and genre ($\beta_1 = 1.240$, $p < 0.05$) were likely to have more revenue than the movies with no similarity, supporting Hypothesis 3. Similarly, tone matching ($\beta_1 = 0.302$, $p < 0.01$) also showed a positive significant effect, even though the size of the coefficient was smaller, supporting Hypothesis 4. We further examined AIC and BIC for

Table 6

Linear regression with interaction terms.

VARIABLES	(1) lnFirstRevenue	(2) lnFirstRevenue
Mood&Genre	−0.653** (0.282)	
Tone		−0.151*** (0.044)
Mood&Genre x MajorStudio	1.240** (0.481)	
Tone x MajorStudio		0.302*** (0.093)
MajorStudio	0.098 (0.114)	0.160* (0.083)
Constant	5.185*** (0.179)	5.178*** (0.178)
lnScreen	YES	YES
lnActorPower	YES	YES
CriticRating	YES	YES
lnCriticCount	YES	YES
US	YES	YES
Foreign	YES	YES
Genre	YES	YES
Year	YES	YES
Months	YES	YES
ViewingRating	YES	YES
Observations	2,851	2,866
R-squared	0.908	0.907

Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.10.

models, demonstrating that the model with Mood&Genre has a better fit than without it.³ The impact of mood or tone dissonance varies depending on whether the movie is produced by a major studio or not.

4.3. Robustness check

4.3.1. Validation with alternative dependent variable

Following our main estimation, we conducted various robustness checks to ensure the reliability of our results. We investigated the possibility of multicollinearity by calculating the variance inflation factors (VIFs) for each variable in our study using simple OLS regression methods. The highest VIF recorded was 7.61 for the variable ‘Drama’ within the Genre category, with an overall average VIF of 2.46. These values were under the generally accepted limit of 10 (Kennedy, 2003, p. 213), indicating that multicollinearity seems not to be of great concern.

We also checked whether the effects on the box office revenue for the opening week still hold on the total box office revenue with the following equations:

$$\begin{aligned} \ln \text{Revenue}_i = & \beta_0 + \beta_1 \text{Mood\&Genre}_i + \beta_2 \ln \text{MaxScreen}_i + \beta_3 \ln \text{ActorPower}_i \\ & + \beta_4 \text{ReviewRating}_i + \beta_5 \ln \text{ReviewCount}_i + \beta_6 \text{CriticRating}_i \\ & + \beta_7 \ln \text{CriticCount}_i + \beta_8 \text{Country}_i + \beta_9 \text{Genre}_i + \beta_{10} \text{Year}_i \\ & + \beta_{11} \text{Month}_i + \beta_{12} \text{MajorStudio}_i + \beta_{13} \text{ViewingRating}_i + \varepsilon_i \end{aligned} \quad (4)$$

$$\begin{aligned} \ln \text{Revenue}_i = & \beta_0 + \beta_1 \text{Tone}_i + \beta_2 \ln \text{MaxScreen}_i + \beta_3 \ln \text{ActorPower}_i \\ & + \beta_4 \text{ReviewRating}_i + \beta_5 \ln \text{ReviewCount}_i + \beta_6 \text{CriticRating}_i \\ & + \beta_7 \ln \text{CriticCount}_i + \beta_8 \text{Country}_i + \beta_9 \text{Genre}_i + \beta_{10} \text{Year}_i \\ & + \beta_{11} \text{Month}_i + \beta_{12} \text{MajorStudio}_i + \beta_{13} \text{ViewingRating}_i + \varepsilon_i \end{aligned} \quad (5)$$

³ The model with Mood&Genre has a lower AIC (63.194 difference) and BIC (51.472 difference) compared to the model without Mood&Genre, indicating a better fit. Similarly, the model with Tone shows a substantial difference. Both AIC and BIC differences exceed 10, which Burnham and Anderson (2004) consider significant. This confirms that Model including Mood&Genre is statistically preferable to Model excluding the variable.

Table 7

Linear regression on the total box office revenue.

VARIABLES	(1) lnRevenue	(2) lnRevenue	(3) lnRevenue	(4) lnRevenue	(5) lnRevenue
Mood&Genre	−0.582*** (0.222)		−0.665*** (0.181)		−0.752*** (0.194)
Tone		−0.083*** (0.035)		−0.085*** (0.029)	−0.107*** (0.032)
Mood&Genre x MajorStudio					0.882** (0.345)
Tone x MajorStudio					0.237*** (0.069)
lnFirstRevenue			0.445*** (0.017)	0.446*** (0.017)	0.443*** (0.017)
lnMaxScreen	1.251*** (0.012)	1.253*** (0.012)	0.717*** (0.023)	0.716*** (0.024)	0.718*** (0.023)
lnActorPower	0.016*** (0.003)	0.016*** (0.003)	0.016*** (0.003)	0.016*** (0.003)	0.016*** (0.003)
MajorStudio	0.358*** (0.055)	0.355*** (0.055)	−0.057 (0.048)	−0.062 (0.048)	−0.368*** (0.099)
ReviewRating	0.093*** (0.024)	0.095*** (0.024)	0.089*** (0.019)	0.090*** (0.018)	0.087*** (0.019)
lnReviewCount	−0.005 (0.011)	0.006 (0.011)	−0.001 (0.009)	−0.001 (0.009)	0.001 (0.009)
CriticRating	0.013*** (0.001)	0.013*** (0.001)	0.014*** (0.001)	0.014*** (0.001)	0.014*** (0.001)
lnCriticCount	0.128*** (0.023)	0.132*** (0.024)	0.139*** (0.019)	0.142*** (0.019)	0.143*** (0.019)
US	0.002 (0.055)	0.004 (0.055)	−0.050 (0.049)	−0.046 (0.049)	−0.048 (0.049)
Foreign	0.092 (0.061)	0.094 (0.061)	0.103* (0.053)	0.103* (0.053)	0.096* (0.053)
Constant	5.722*** (0.196)	5.739*** (0.195)	2.822*** (0.201)	2.839*** (0.200)	2.907*** (0.202)
Genre	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
Months	YES	YES	YES	YES	YES
ViewingRating	YES	YES	YES	YES	YES
Observations	2,851	2,866	2,851	2,866	2,851
R-squared	0.932	0.932	0.953	0.953	0.953

Robust standard errors in parentheses.

*** p < 0.01, ** p < 0.05, * p < 0.10.

Column (1) in Table 7 shows the result of regression on the total box office revenue with Mood&Genre. The coefficient of the similarity between poster mood and genre ($\beta_1 = -0.582$, $p < 0.01$) was significantly negative. Similarly, Column (2) in Table 7 shows that the coefficient of tone similarity was also significantly negative. The effects of the variables were the same as the main results.

We also considered the possibility of the impact of opening week revenues on the total revenues with the following equations:

$$\begin{aligned} \ln \text{Revenue}_i = & \beta_0 + \beta_1 \text{Mood\&Genre}_i + \beta_2 \ln \text{FirstRevenue}_i + \beta_3 \ln \text{MaxScreen}_i \\ & + \beta_4 \ln \text{ActorPower}_i + \beta_5 \text{ReviewRating}_i + \beta_6 \ln \text{ReviewCount}_i \\ & + \beta_7 \text{CriticRating}_i + \beta_8 \ln \text{CriticCount}_i + \beta_9 \text{Country}_i \\ & + \beta_{10} \text{Genre}_i + \beta_{11} \text{Year}_i + \beta_{12} \text{Month}_i + \beta_{13} \text{MajorStudio}_i \\ & + \beta_{14} \text{ViewingRating}_i + \varepsilon_i \end{aligned} \quad (6)$$

$$\begin{aligned} \ln \text{Revenue}_i = & \beta_0 + \beta_1 \text{Tone}_i + \beta_2 \ln \text{FirstRevenue}_i + \beta_3 \ln \text{MaxScreen}_i \\ & + \beta_4 \ln \text{ActorPower}_i + \beta_5 \text{ReviewRating}_i + \beta_6 \ln \text{ReviewCount}_i \\ & + \beta_7 \text{CriticRating}_i + \beta_8 \ln \text{CriticCount}_i + \beta_9 \text{Country}_i \\ & + \beta_{10} \text{Genre}_i + \beta_{11} \text{Year}_i + \beta_{12} \text{Month}_i + \beta_{13} \text{MajorStudio}_i \\ & + \beta_{14} \text{ViewingRating}_i + \varepsilon_i \end{aligned} \quad (7)$$

$$\begin{aligned} \ln \text{Revenue}_i = & \beta_0 + \beta_1 \text{Mood\&Genre}_i + \beta_2 \text{Tone}_i + \beta_3 \text{Mood\&Genre}_i \\ & \times \text{MajorStudio}_i + \beta_4 \text{Tone}_i \\ & \times \text{MajorStudio}_i + \beta_5 \text{MajorStudio}_i + \beta_6 \ln \text{FirstRevenue}_i \\ & + \beta_7 \ln \text{MaxScreen}_i + \beta_8 \ln \text{ActorPower}_i + \beta_9 \text{ReviewRating}_i \\ & + \beta_{10} \ln \text{ReviewCount}_i + \beta_{11} \text{CriticRating}_i + \beta_{12} \ln \text{CriticCount}_i \end{aligned}$$

$$\begin{aligned} & + \beta_{13} \text{Country}_i + \beta_{14} \text{Genre}_i + \beta_{15} \text{Year}_i + \beta_{16} \text{Month}_i \\ & + \beta_{17} \text{ViewingRating}_i + \varepsilon_i \end{aligned} \quad (8)$$

The number of people watching movies in the opening week has a chance to influence other people to choose the movie due to word of mouth. To maintain consistency in the effects of mood and tone, Columns (3) and (4) incorporate the same variables as those in Columns (1) and (2). Additionally, we included the variable of opening week revenue in Columns (3), (4) and (5) for testing purposes. We observed consistent results (Mood&Genre in Column (3): $\beta_1 = -0.665$, $p < 0.01$; Tone in Column (4): $\beta_1 = -0.085$, $p < 0.01$; Mood&Genre x MajorStudio in Column (5): $\beta_1 = 0.882$, $p < 0.05$; Tone x MajorStudio in Column (5): $\beta_1 = 0.237$, $p < 0.01$).

4.3.2. Validation with online survey

To examine expectancy violations, we conducted an online experiment with 794 participants (50.25 % male, average age: 41.35 years), who successfully passed an attention check. This study was conducted using Connct by CloudResearch and utilized a 2 (High Similarity vs. Low Similarity) $\times$ 8 (Genres) between-subjects design. The order of the movie poster and genre were randomly shown in the survey. The questions assessing expectancy violations were modified from prior literature (Bevan et al., 2014; Spreng et al., 1996). Participants evaluated whether the poster and genre not aligned with their expectations and to rate the degree of difference on a seven-point scale. Then, we asked how curious do they feel about the movie (Menon & Soman, 2002). The results are summarized in Table 8, showing that low similarity between movie poster and genre creates expectancy violations

Table 8
Results of linear regression.

VARIABLES	(1) Violation	(2) Difference	(3) Curiosity
Lowsimilarity	0.447*** (0.135)	0.768*** (0.134)	0.231* (0.134)
Age	YES	YES	YES
Gender	YES	YES	YES
HowOften	YES	YES	YES
Household Income	YES	YES	YES
Household	YES	YES	YES
Education	YES	YES	YES
Watched	YES	YES	YES
Observations	794	794	794

Notes: Robust standard errors in parentheses. The intercept is omitted from the table.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(Column (1): $\beta = 0.447$, $t = 3.30$, $p < 0.001$). The participants also perceived greater differences as similarity decreased (Column (2): $\beta = 0.768$, $t = 5.74$, $p < 0.001$). There is some evidence that low similarity created curiosity (Column (3): $\beta = 0.231$, $t = 1.72$, $p < 0.1$). The results revealed that low similarity between movie posters and their genre triggered expectancy violations, which in turn sparked curiosity, regardless of whether the expectation was initially formed from the poster or the genre. The participants also perceived greater differences as similarity decreased, confirming the validity of the similarity measure. Therefore, this finding validates the approach used in the main studies.

5. Discussion

5.1. Conclusion

This research offers novel insights into the strategic role of movie posters as a promotional tool, emphasizing how visual design can shape consumer decisions. To the best of our knowledge, this is the first study to introduce variables that measure the similarity of elements between movie posters and the information released in movie summaries such as genres.

The findings suggest that if the mood conveyed by movie posters deviates from the expected genre norms, it captivates the potential audience's curiosity and drives higher box office revenue. A similar effect applies to the poster's tone, which is another factor contributing emotional impact of images. A negative relationship between the tone of movie posters and the movie's synopsis suggests that dissonance in tone also boosts opening week revenues. These findings imply that strategic contrasts in mood and tone in movie posters can attract more audiences, possibly due to the intrigue or curiosity such disparities invoke.

Overall, these counterintuitive results align with the concept of expectancy violations theory. Moreover, we also validated these findings through an online experiment. The results found that low similarity between movie posters and their genre caused expectancy violations, leading to curiosity regardless of whether the expectation originated from the poster or the genre. Potential customers have expectations about a movie, and when these expectations are violated, it can lead to significant outcomes. This suggests that the impact of visual discrepancies plays a crucial role in consumer engagement.

Additionally, we compared results based on two types of film studios: major and non-major. Notably, for non-major studios, a high degree of similarity between the poster's mood and the movie's genre corresponded to lower revenues, indicating that differentiation and niche appeal are more beneficial for smaller studios. Non-major studios may find success through innovative storytelling and genre-blending, while major studios can capitalize on genre conventions and cohesive production values to attract broader audiences. Therefore, all analyses consistently demonstrate the significance of the contrasting moods

portrayed in movie posters, particularly for films produced by non-major studios.

5.2. Theoretical contributions

This research generates three theoretical contributions. First, this study extends the literature by introducing new variables that have an impact on movie revenue: mood and tone similarities between movie posters and released information. In studies of box office performance, scholars have investigated numerous features of movies to predict their box office success, given the high level of uncertainty in the film industry due to limited information (Elberse, 2007; Bae & Kim, 2019; Zhou et al., 2019; Lampel & Shamsie, 2000; Wi et al., 2020; Chao et al., 2023). Therefore, a new variable configuration is crucial in such studies. Our findings demonstrate that unexpected twists in movie posters significantly influence potential customers' decision-making processes. This is the first study to provide the importance of such variables in box office performance.

Second, this study makes an extension to prior research on the expectancy violations theory. In this study, we demonstrate that expectancy violations of movie posters can intrigue audiences, potentially boosting box office revenues, which refers to positive expectancy violations. Furthermore, this research empirically presents the effect of expectancy violation of movie posters, specifically in non-major studios. Non-major studios can benefit from differentiation and twist, suggesting a niche marketing strategy that leverages novelty.

Lastly, this research provides a state-of-the-art method for emotional representations of images, which is a significant enhancement in marketing literature. Previously, research examining image mood predominantly relied on survey-based experiments. In contrast, this study utilized an extensive collection of images and computer vision technology, such as Gemini, to analyze image mood, thereby suggesting novel applications for the mood conveyed by movie posters. This bridges theory with advancements in AI, paving the way for future investigations into consumer psychology and marketing strategies in the digital age.

5.3. Practical and managerial implications

This study offers several insights for marketing managers in the film industry, particularly in relation to the strategic deployment of artificial intelligence (AI) and visual design in promotional materials.

First, this study highlights the incorporation of artificial intelligence (AI) in enhancing box office performance. It reveals that movie studios can significantly benefit from employing innovative marketing tactics, such as designing movie posters that deliberately contrast with the actual content or genre of the movie. This approach not only generates curiosity among potential viewers but also encourages a departure from traditional marketing practices, favoring curiosity-driven campaigns. Additionally, the importance of differentiation through contrast mood in posters is highlighted, particularly for non-major studios aiming to carve out niche markets. This strategy enables them appeal to audiences seeking novel and plot twist movie experiences.

Second, another key practical implication is that the mood conveyed through images is a manageable component of the marketing strategy. This insight highlights the significant power managers possess in shaping audience perceptions and engagement through the strategic use of visual promotional materials. By thoughtfully curating the mood of these images, such as movie posters, managers can subtly guide audience expectations and reactions. Whether aligning these visual cues closely with the expected mood of the movie or intentionally creating a contrast to spark curiosity, the ability to control the narrative without relying on unpredictable reviews or critics becomes a crucial advantage. Consequently, this study offers marketing managers in the film industry a powerful tool to navigate the complexities of consumer engagement, emphasizing the strategic importance of image mood manipulation

alongside other marketing efforts.

Finally, integrating Generative AI (Gen-AI) into marketing analysis and prediction tools enhances the ability to anticipate consumer preferences and potential market success of upcoming movies. By systematically analyzing and designing promotional materials that evoke the intended emotional responses, studios, and marketing agencies can optimize their strategies for maximum impact. Moreover, the use of Gen-AI in predictive analytics significantly enhances the ability to forecast box office performance, allowing for more informed decisions regarding promotional efforts. This technological integration represents a significant leap forward in leveraging data-driven insights to tailor marketing approaches and achieve greater market success.

5.4. Limitations and future research directions

This research sheds light on the strategic importance of image mood in marketing within the film industry but encounters certain limitations that suggest avenues for future exploration.

First, the study’s focus on the film sector limits its applicability to other industries where visual marketing is crucial. Expanding this research to various sectors could validate the findings’ universality and effectiveness across different contexts.

Second, the impact of cultural differences on the perception of image mood was not deeply explored. Our data included only a small number of movies released internationally. This is likely due to the limited number of distributors releasing blockbuster films internationally. This data limitation may cause bias into the dataset, making it unsuitable for a cross-cultural analysis. As an alternative, we further investigated the interaction effects between mood and movie genres or country, however the data in this study did not yield significant results. Future research could benefit from cross-cultural studies to tailor marketing strategies effectively to diverse global audiences.

Third, the study utilized opening week revenue as a measure to assess the impact of the movie poster before the public feedback or broader audience reactions could influence outcomes. While consistent results were observed when regressions included total revenue and on-line review-related variables, opening week viewership can still be

influenced by other factors in mass media. Although our approach isolates the impact of expectancy violations before extensive audience feedback emerges, this short-term focus does not account for potential negative repercussions over the entire theatrical run or in downstream markets. Thus, these findings do not necessarily imply that expectancy violation improves movie evaluations. A more comprehensive understanding of expectancy violation should therefore consider both immediate effects and possible aftereffects.

Lastly, exploring the realm of video content through the use of generative AI and large language models marks a promising extension of current research into visual marketing’s impact. This approach particularly applied to the analysis of movie trailers, allows for an in-depth examination of how dynamic elements like scene composition, pacing, and sound influence consumer behavior. By leveraging advanced AI to dissect these aspects, researchers can uncover predictive patterns of audience engagement, offering a richer understanding of video marketing’s effectiveness. This progression not only widens the scope of the investigation beyond static images but also aligns with contemporary shifts toward digital media, providing advertising practitioners with insights for optimizing video marketing strategies to captivate and engage audiences more effectively.

CRediT authorship contribution statement

Youngjun Kim: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Hye-Jin Kim:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Keeyeon Kicheon Park:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

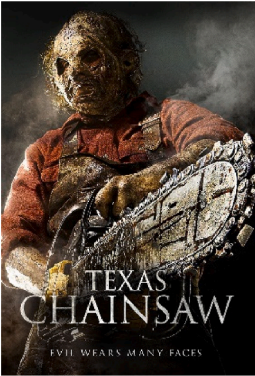
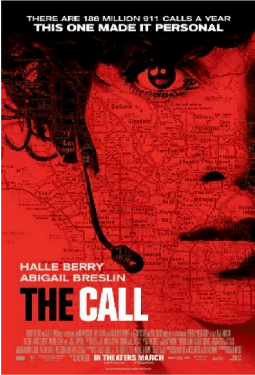
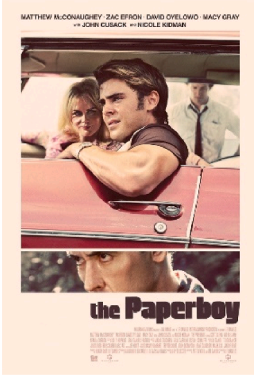
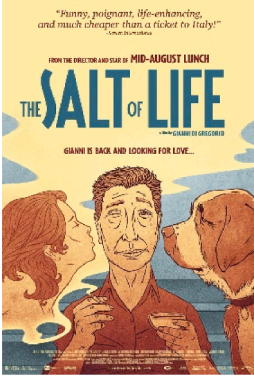
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. . Examples of movie posters

High Similarity		Low Similarity	
	TitleCaptain America Civil War GenreAction MoodIntenseSuspensefulEpicClimacticConfrontational Mood&Genre0.173 ToneDark		TitleHimmatwala GenreAction MoodColorfulVibrantEnergeticFestiveLively Mood&Genre0.091 ToneBright

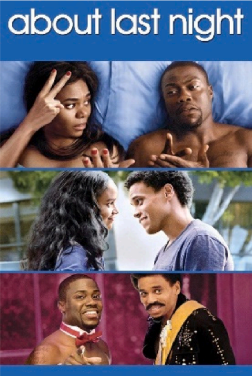
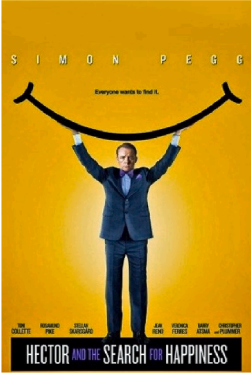
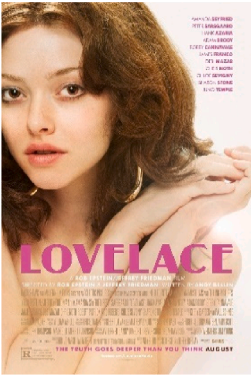
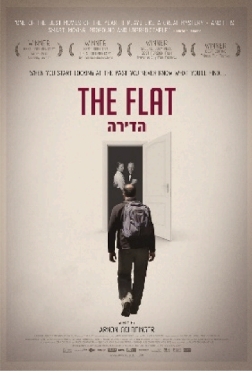
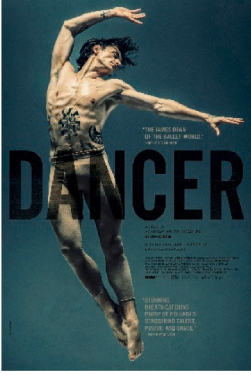
(continued on next page)

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High Similarity			Low Similarity		
	Title Genre Mood	Texas Chainsaw Horror Scary Creepy Disturbing Horrific Terrifying		Title Genre Mood	The Lazarus Effect Horror Angry Dangerous Disturbed Menacing Terrifying
	Mood&Genre Tone	0.462 Dark		Mood&Genre Tone	0.232 Dark
	Title Genre Mood	The Call Thriller/Suspense Mysterious Suspenseful Dark Gritty Edgy		Title Genre Mood	The Paperboy Thriller/Suspense Long Thoughtful Calm Curious Happy
	Mood&Genre Tone	0.332 Dark		Mood&Genre Tone	0.101 Bright
	Title Genre Mood	Identity Thief Comedy Awkward Funny Quirky Romantic Heartwarming		Title Genre Mood	The Salt of Life Comedy Longed Desired Expected Awaited Hoped-for
	Mood&Genre Tone	0.351 Bright		Mood&Genre Tone	0.008 Dark

(continued on next page)

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High Similarity			Low Similarity		
	Title	Rogue One: A Star Wars Story		Title	The Revenant
	Genre	Adventure		Genre	Adventure
	Mood	Epic		Mood	Intense
		Adventurous			Vulnerable
		Thrilling			Defiant
		Suspenseful			Determined
		Exciting			Focused
	Mood&Genre	0.431		Mood&Genre	0.056
	Tone	Dark		Tone	Dark
	Title	About Last Night		Title	Hector And The Search For Happiness
	Genre	Romantic/Comedy		Genre	Romantic/Comedy
	Mood	Funny		Mood	Powerful
		Romantic			Hopeful
		Heartwarming			Optimistic
		Sexy			Determined
		Feel-good			Confident
	Mood&Genre	0.433		Mood&Genre	0.046
	Tone	Bright		Tone	Dark
	Title	Son of Saul		Title	Lovelace
	Genre	Drama		Genre	Drama
	Mood	Intense		Mood	Long
		Dramatic			Soft
		Suspenseful			Warm
		Emotional			Inviting
		Gritty			Vulnerable
	Mood&Genre	0.347		Mood&Genre	0.014
	Tone	Dark		Tone	Bright
	Title	The Flat		Title	Dancer
	Genre	Documentary		Genre	Documentary
	Mood	Mysterious		Mood	Graceful
		Eerie			Elegant
		Suspenseful			Poised
		Intriguing			Lithe
		Haunting			Athletic
	Mood&Genre	0.207		Mood&Genre	0.004
	Tone	Dark		Tone	Dark

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